API Documentation

Student Success Assistant REST Microservice Specification

1. API Overview & Authentication

The Student Success Assistant exposes a high-performance FastAPI REST API. All endpoints (except /api/health) require authentication via the X-API-Key request header. Administrative endpoints (such as applying feedback loop updates) require the X-Admin-Key header. Interactive OpenAPI documentation is hosted at /api/docs.

Base URL: http://127.0.0.1:8000
Standard Headers:
• X-API-Key: <STUDENT_ASSISTANT_API_KEY>
• Content-Type: application/json

2. Endpoint Catalog & Specifications

POST /api/analyze

Executes full multi-agent student success analysis, risk diagnosis, and intervention planning.

Sample Request (JSON)	Sample Response (JSON)

<pre>{
  "student": {
  "id_student": 59272,
  "code_module": "BBB",
  "code_presentation": "2014J"
  },
  "checkpoint": 0.40
}</pre>
	<pre>{
	  "student": {"id_student": 59272, "code_module": "BBB", "code_presentation": "2014J"},
	  "checkpoint": 0.40,
	  "withdrawal_risk": {"withdrawal_risk": 0.68, "risk_tier": "High"},
	  "plan": {
	  "narrative": {"summary": "Student has fallen behind on submission deadlines..."},
	  "actions": [{"action_id": "tutor_check_in", "title": "1-on-1 Academic Tutoring"}],
	  "critic_report": {"valid": true, "violations": []}
	  }
	}</pre>

POST /api/predict-risk

Predicts calibrated withdrawal probability, risk tier, and TreeSHAP feature drivers.

Sample Request (JSON)	Sample Response (JSON)

```
{
  "student": {
    "id_student": 59272,
    "code_module": "BBB",
    "code_presentation": "2014J"
  },
  "checkpoint": 0.40
}
```

```
{
  "withdrawal_risk": {
    "withdrawal_risk": 0.682,
    "risk_tier": "High",
    "drivers": [
      {"label": "Days since last activity", "direction": "increases_risk", "shap_value": 0.412},
      {"label": "Missed assessments", "direction": "increases_risk", "shap_value": 0.285}
    ]
  },
  "outcome_prediction": {
    "predicted_outcome": "Fail",
    "p_poor_outcome": 0.814
  }
}
```


POST /api/retrieve-similar

Hybrid RAG search for similar historical student precedents with 'what happened next' stories.

Sample Request (JSON)	Sample Response (JSON)


```
{
  "query": "repeat students with heavy credits and low engagement who withdrew",
  "top_k": 5,
  "checkpoint": 0.40,
  "filters": {"code_module": "BBB", "num_of_prev_attempts": {"min": 1}}
}
```

```
{
  "outcome_summary": {
    "withdrawal_rate": 0.72,
    "poor_outcome_rate": 0.88,
    "withdrawal_rate_95ci": {"lower": 0.58, "upper": 0.83}
  },
  "cases": [
    {
      "case_id": "BBB_2013J_48123",
      "final_result": "Withdrawn",
      "score": 0.891,
      "story": {"text": "Student was inactive for 3 weeks after TMA 01 and withdrew in week 18."}
    }
  ]
}
```


POST /api/prioritize

Ranks students within a cohort according to capacity-optimized risk weights.

Sample Request (JSON)	Sample Response (JSON)

<table>
<tr>
<td>

```
{
  "code_module": "BBB",
  "code_presentation": "2014J",
  "capacity": 0.20,
  "limit": 10,
  "checkpoint": 0.40
}
```

</td>
<td>

```
{
  "cohort_size": 1240,
  "capacity": 0.20,
  "ranking": [
    {
      "priority_rank": 1,
      "id_student": 59272,
      "risk_tier": "High",
      "withdrawal_risk": 0.682,
      "reasons": ["Inactive last 18 days", "Missed TMA 02
"]
    }
  ]
}
```

</td>
</tr>
</table>

POST /api/copilot

Conversational assistant with function-calling over 8 grounded analytical tools.

Sample Request (JSON)	Sample Response (JSON)

<pre>{
  "question": "Who should I contact first in BBB 2014J an
d why?",
  "checkpoint": 0.40
}</pre>
	<pre>{
	  "answer": "In BBB 2014J, student 59272 has the highest
	priority rank due to 18 days of zero VLE activity and mis
	sing TMA 02.",
	  "tool_calls": [
	    {"tool": "rank_cohort", "arguments": {"code_module":
	"BBB", "code_presentation": "2014J", "capacity": 0.1}}
	  ],
	  "grounding_verified": true
	}</pre>

POST /api/feedback

Records advisor feedback on recommended plans, accepted actions, and observed semester outcomes.

Sample Request (JSON)	Sample Response (JSON)

<pre>{
  "recommendation_id": 14,
  "rating": 5,
  "accepted": true,
  "notes": "Tutor outreach was successful; student comple
ted TMA 02.",
  "observed_outcome": "Pass"
}</pre>
	<pre>{
	  "status": "recorded",
	  "feedback_id": 82
	}</pre>